

Supplementary Materials

Pupil stuff

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Priors

These are the priors used for the preregistered LBA model.

Table 1. Priors of the preregistered linear ballistic accumulator model.

Parameter	Description	Condition	Correct	Response	Mean	SD
v_SNovel:IMFALSE	Drift rate	Novel	False	-	0	2
v_SLearned:IMFALSE	Drift rate	Learned	False	-	0	2
v_SNovel:IMTRUE	Drift rate	Novel	True	-	1	2
v_SLearned:IMTRUE	Drift rate	Learned	True	-	1	2
sv_IMTRUE	SD drift rate (true accumulator)	-	True	-	$\log(0.5)$	0.5
B_IRNovel	Threshold	-	-	Novel	$\log(1)$	0.5
B_IRLearned	Threshold	-	-	Learned	$\log(1)$	0.5
A	Start point	-	-	-	$\log(0.5)$	0.3
t0	Non-decision time	-	-	-	$\log(0.3)$	0.3

Model checks

Convergence

The chains showed satisfactory mixing as evidenced by the rhat values.

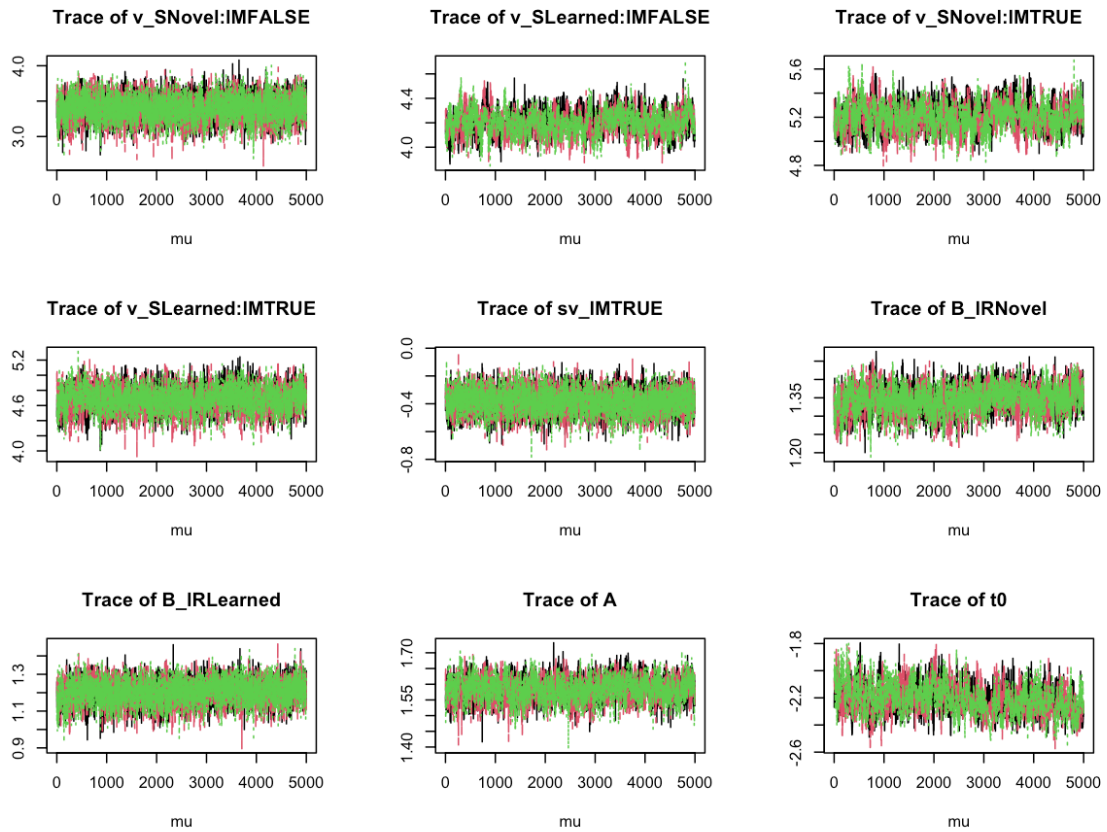
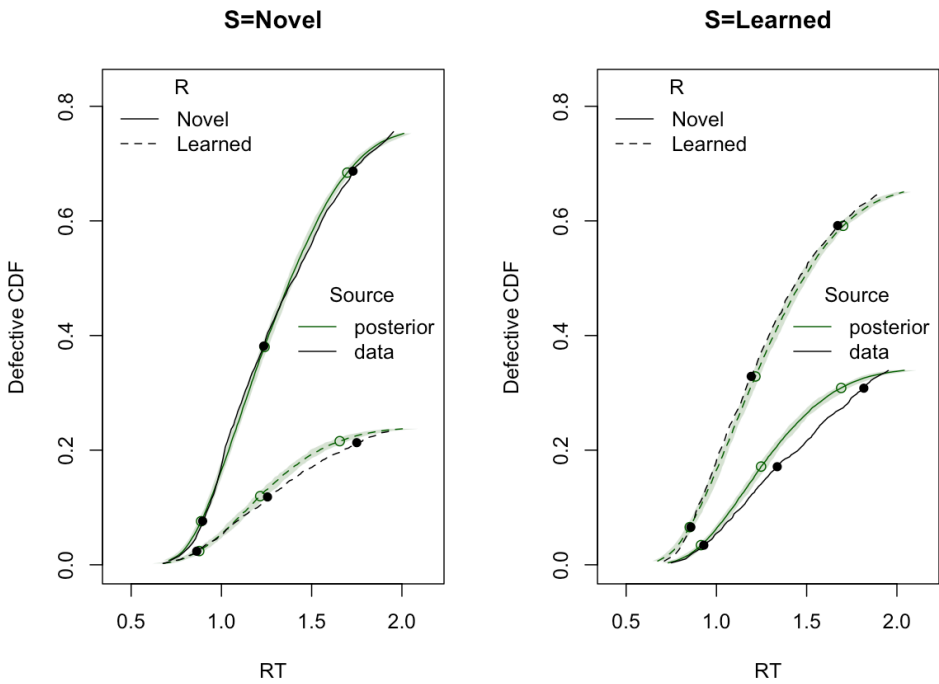
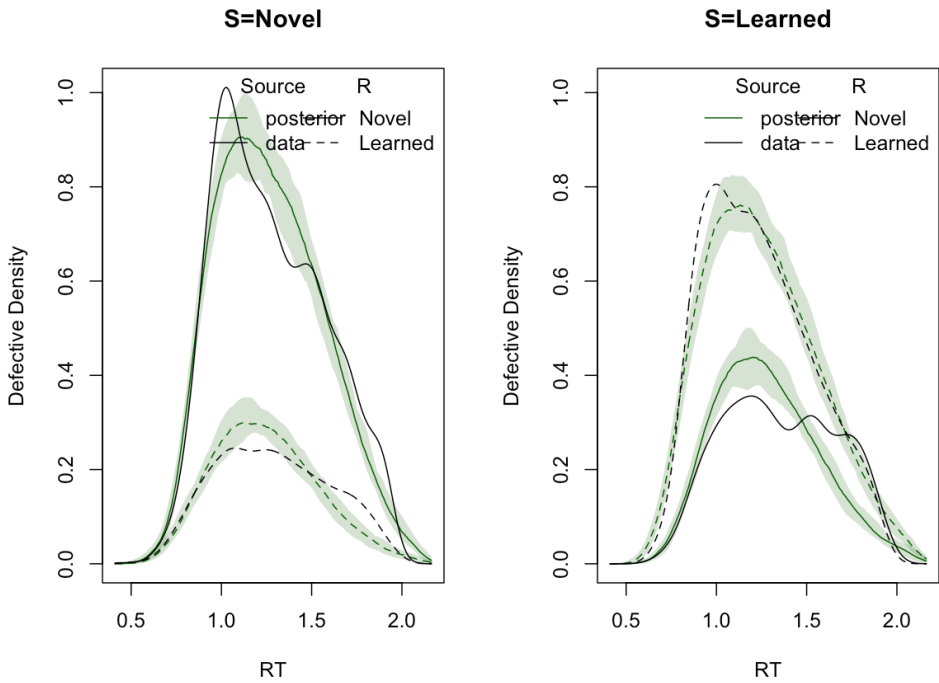


Figure 1. MCMC chains

Posterior predictive checks



(a) CDF



(b) Density

Figure 2. Posterior predictive checks

Prior vs posterior plots

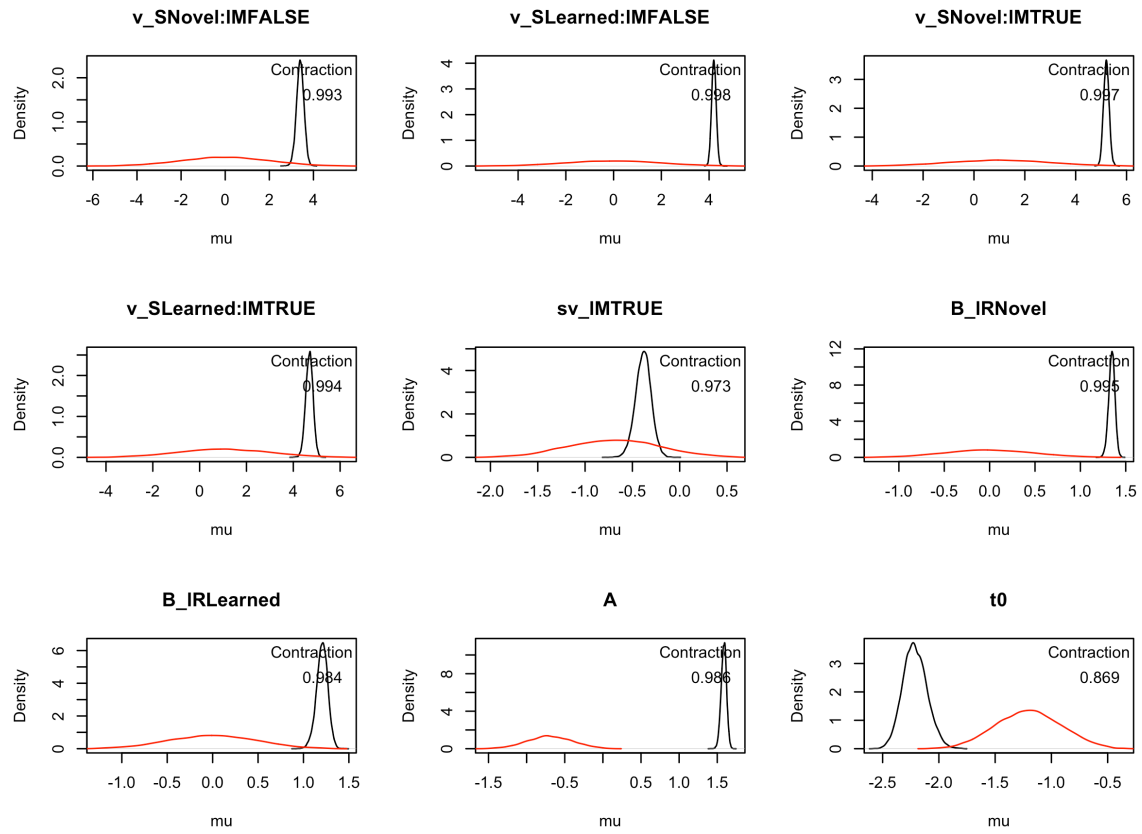
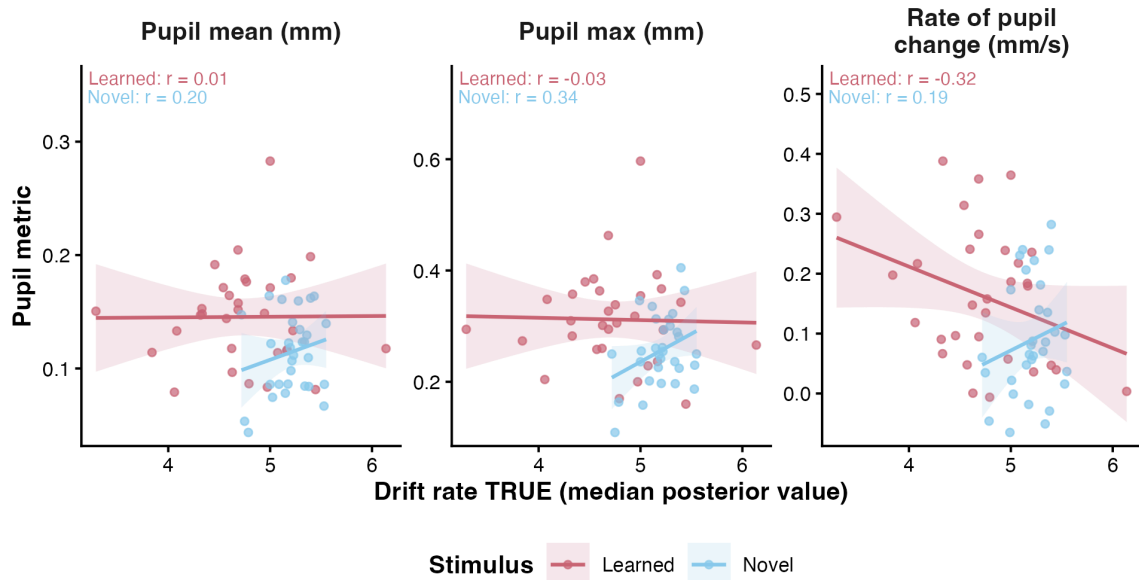


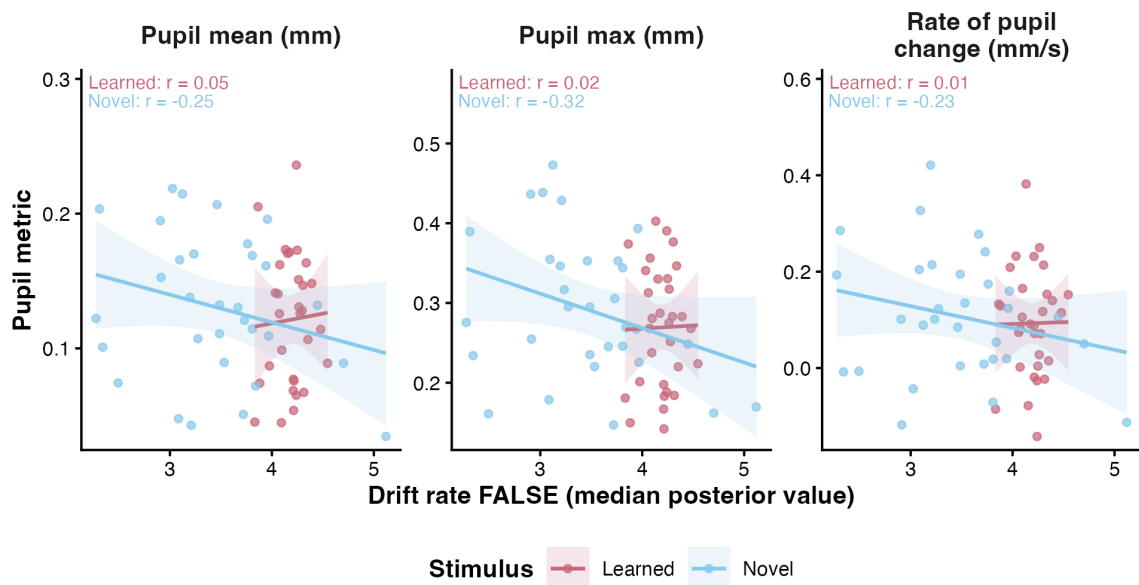
Figure 3. Priors vs posterior plots

Point estimate correlations

Linear relationships between the median of the posterior distribution of drift rate (TRUE) and drift rate (FALSE) and the three pupil metrics (mean pupil size, peak pupil size and rate of pupil change) can be seen in Figure 4.



(a) Point estimates correlations between drift rate (TRUE) for Learned and Novel faces and pupil metrics (mean pupil size, peak pupil size and rate of pupil change)

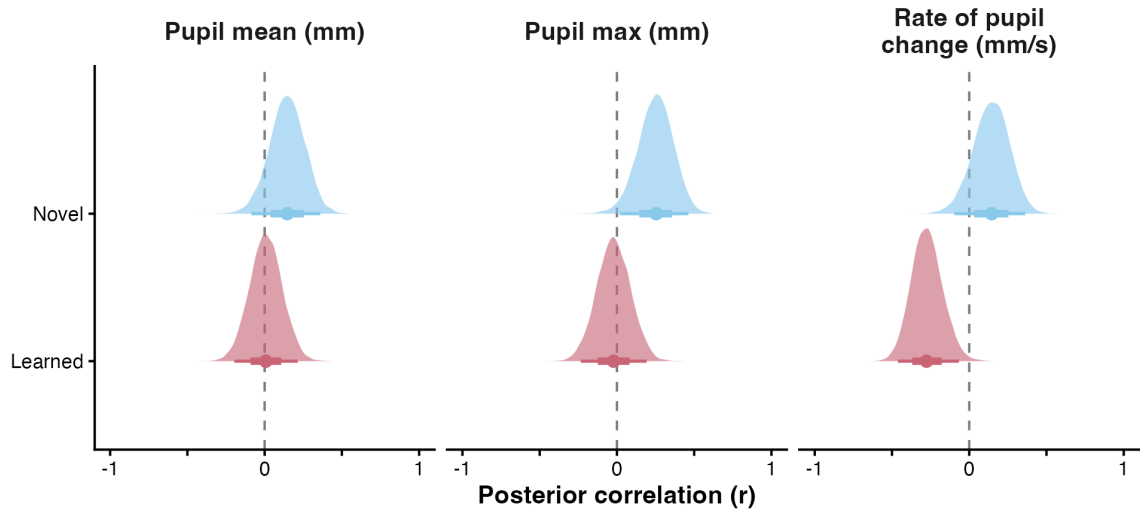


(b) Point estimates correlations between drift rate (FALSE) for Learned and Novel faces and pupil metrics (mean pupil size, peak pupil size and rate of pupil change)

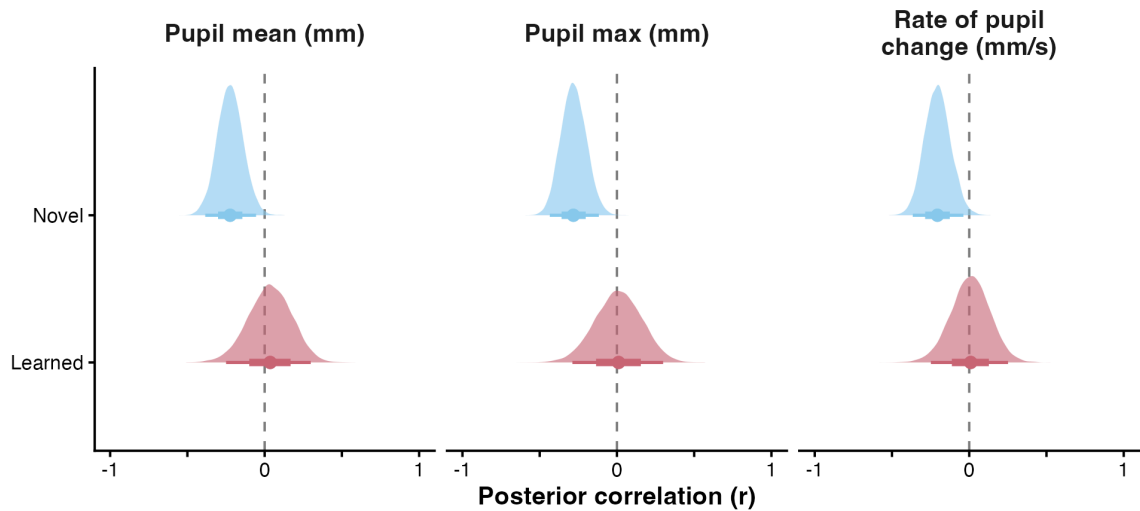
Figure 4. Point estimates correlations between drift rate (TRUE) and drift rate (FALSE) and pupil metrics

Plausible value correlations

We calculated the plausible value correlations between each posterior draw of drift rate (TRUE) and drift rate (FALSE) and the three pupil metrics (mean pupil size, peak pupil size and rate of pupil change). These distributions can be seen in Figure 5 and a summary of these distributions can be seen in Table 2.



(a) Plausible value correlations between drift rate (TRUE) for Learned and Novel faces and pupil metrics (mean pupil size, peak pupil size and rate of pupil change)



(b) Plausible value correlations between drift rate (FALSE) for Learned and Novel faces and pupil metrics (mean pupil size, peak pupil size and rate of pupil change)

Figure 5. Plausible value correlations between drift rate parameters and pupil metrics. Points denote the median of the distribution and error bars represent the 66% and 95% quantile intervals.

Table 2. Median r and 95% quantile intervals of the plausible value correlations between drift rate (TRUE and FALSE) and pupil metrics.

Parameter	Condition	Accuracy	Pupil mean		Pupil max		Pupil change rate	
			Median r	95% QI	Median r	95% QI	Median r	95% QI
Drift	Learned	TRUE	0.01	[-0.20, 0.21]	-0.02	[-0.23, 0.19]	-0.28	[-0.46, -0.07]
Drift	Learned	FALSE	0.04	[-0.25, 0.30]	0.01	[-0.29, 0.30]	0.01	[-0.25, 0.25]
Drift	Novel	TRUE	0.15	[-0.09, 0.36]	0.25	[0.02, 0.46]	0.15	[-0.10, 0.36]
Drift	Novel	FALSE	-0.22	[-0.38, -0.06]	-0.28	[-0.43, -0.12]	-0.21	[-0.37, -0.04]

Note. Blue = 95% QI entirely above zero; red = 95% QI entirely below zero.

Joint modelling

We calculated the entire correlation matrix from the posterior median draws of the joint model of mean pupil size and LBA parameters. This correlation matrix can be seen in Figure 6.

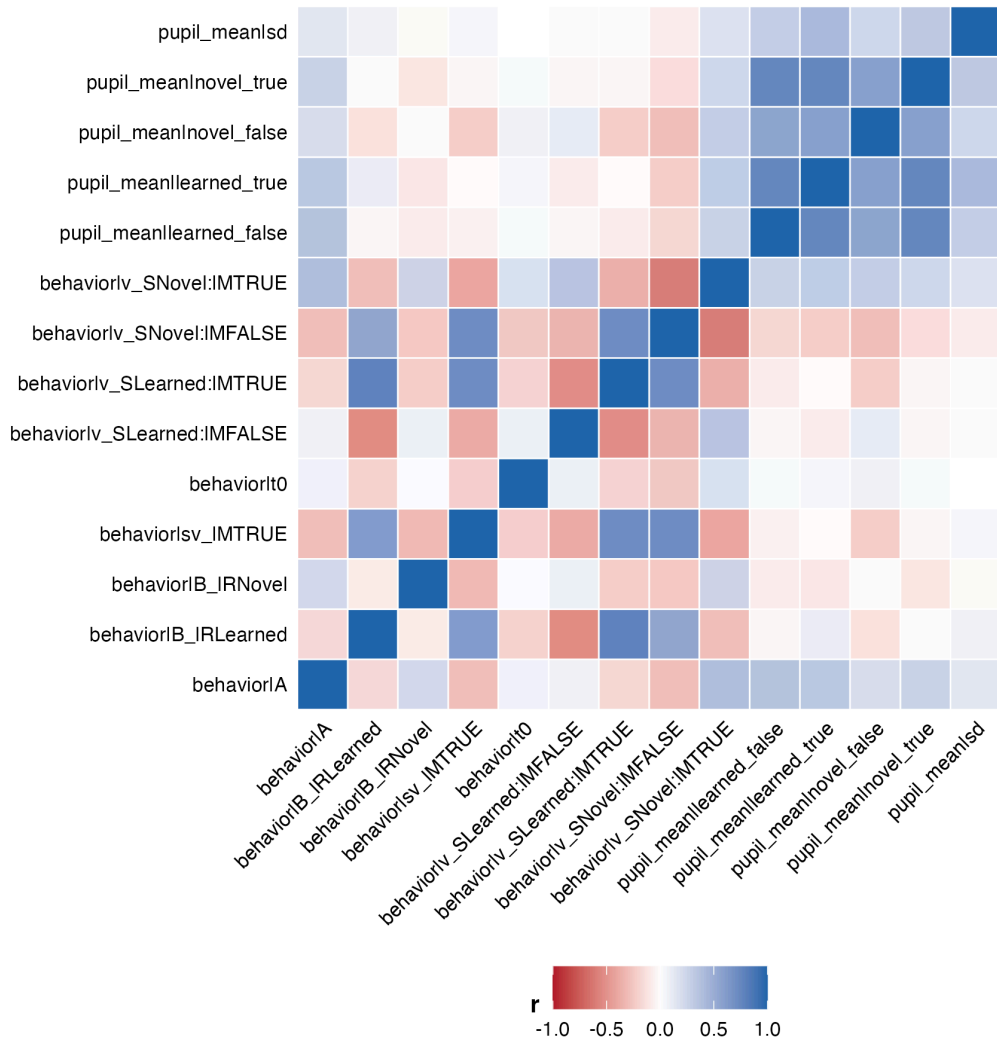


Figure 6. Correlation matrix of the posterior medians of the group-level correlations between LBA and pupil parameters estimated in the joint model. Blue = positive correlation; red = negative correlation.

References